Zeroth Review

Name of the department: ELECTRONICS AND COMMUNICATION ENGINEERING
Course code & Title: U23EC651 Project with Design Thinking

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Brainstorming for Project Titles

PROBLEM STATEMENT

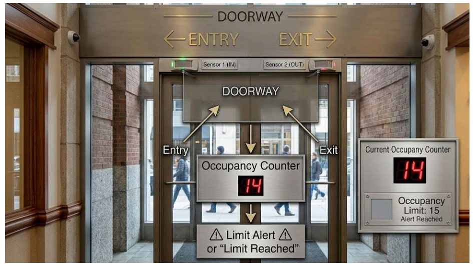
In modern smart buildings, classrooms, offices, and large public venues, maintaining safe occupancy levels is essential for ensuring safety, user comfort, and efficient utilization of resources such as lighting, ventilation, and air-conditioning systems. Overcrowding in confined spaces can lead to serious safety hazards, reduced comfort, and unnecessary energy consumption, especially in environments where occupancy directly influences system control.

Most existing occupancy monitoring systems rely on microcontrollers and software-based processing,

- Sequential Processing Delays
- Higher Power Consumption
- Limited Real-Time Performance
- Inaccuracy during fast movement

Objectives

- Design a real-time occupancy monitoring system
- Achieve accurate bidirectional counting of people
- Ensure low power and efficient operation
- Implement overcrowding detection with predefined limits
- Develop a scalable and reliable solution for multiple locations
- Provide real-time status indication for easy monitoring
- Ensure robust and error-free detection under continuous movement



SDG mapping

Sustainable Development Goals (SDG) addressed (Please tick the applicable area)	
1. No Poverty	
2. Zero Hunger	
3. Good Health and Well-being	
4. Quality Education	
5. Gender Equality	
6. Clean Water and Sanitation	
7. Affordable and Clean Energy	
8. Decent Work and Economic Growth	
9. Industry, Innovation, and Infrastructure	✓
10. Reduced Inequality	
11. Sustainable Cities and Communities	✓
12. Responsible Consumption and Production	
13. Climate Action	

Literature Survey

S. No	TITLE	AUTHOR	NAME OF ORGANIZATION & YEAR	DESCRIPTION
1	Infrared-Based Bidirectional Visitor Counter System	S. M. Metev, V. P. Veiko	IEEE / Embedded Systems, 2018	• Uses IR sensors to detect entry and exit based on beam interruption • Implemented using microcontrollers with sequential processing. • Limited accuracy during fast movement.
2	Arduino-Based Smart Room Occupancy System	A. Kumar, R. Singh	International Journal of Engineering Research, 2019	• Low-cost system using Arduino and IR sensors • Easy implementation and suitable for small applications. • Suffers from processing delays and timing issues.
3	IoT-Based Occupancy Monitoring and Energy Control System	M. S. Hossain, G. Muhammad	IEEE IoT Conference, 2020	• Integrates occupancy detection with IoT for remote monitoring. • Controls lighting and ventilation automatically. • Depends on network connectivity and increases system complexity.
4	8051 Microcontroller-Based Visitor Counter	R. S. Gaonkar	Embedded Systems Journal, 2017	• Uses basic IR sensors with 8051 controller • Simple design for classrooms and offices. • Not suitable for real-time high-speed movement detection.
5	Computer Vision-Based People Counting System	Z. Zhang, Y. Wang	IEEE Transactions / Computer Vision, 2021	• Uses cameras and deep learning for high accuracy. • Works well in crowded environments • High power consumption, cost, and privacy concerns.
6	Smart Building Occupancy Detection System	J. A. Stankovic, I. Lee	Smart Cities Journal, 2022	• Focuses on energy-efficient building management. • Uses multiple sensors for occupancy tracking • Requires complex integration and higher system cost.

Proposed Solution

- Develop a real-time room occupancy monitoring system using hardware-based design
- Use sensor-based detection to identify entry and exit of people
- Ensure accurate bidirectional counting for reliable occupancy tracking
- Provide instant updates without delay for safety-critical environments
- Include capacity monitoring to prevent overcrowding
- Design system to be low power and efficient compared to software-based solutions
- Enable scalable deployment across multiple rooms or locations
- Support centralized monitoring for overall occupancy management

Thank you